Context-Sensitive Pragmatic Classification for Burmese: A 4-Stage Ablation Study with End-to-End Pipeline

Thet Su Sann
Asian Institute of Technology
st126316@ait.asia

Thiri Shin Thant
Asian Institute of Technology
st126018@ait.asia

Hein Min Htet
Asian Institute of Technology
st126459@ait.asia

Tisa Bajracharya
Asian Institute of Technology
st126686@ait.asia

Abstract

We present a system for classifying the pragmatic politeness level of Burmese utterances using a fine-tuned multilingual encoder model. Burmese is a low-resource language in which pragmatic meaning is highly sensitive to social context, speaker hierarchy, and linguistic register. We design a four-stage ablation study to systematically investigate how progressively richer contextual signals affect classification quality. In Stage 1, the model receives only the raw utterance. Stage 2 adds social context and communicative instruction. Stage 3 provides explicit pragmatic metadata (register, power relation, tone) as a text prefix serving as an oracle upper bound. Stage 4 closes the loop by replacing oracle metadata with automatic predictions from three independently trained auxiliary classifiers, forming a fully end-to-end pipeline. Our results show that contextual signals improve macro F1 by +0.057 (Stage 1→2) and oracle metadata adds a further +0.041 (Stage 2→3). However, the end-to-end pipeline incurs a -0.097 drop, demonstrating a significant error propagation cost when metadata is predicted rather than given. We further investigate two improved intermediate classifier architectures a codependent chain and a multi-task model and find that improving intermediate classifier accuracy does not translate to end-to-end gains, revealing that the bottleneck lies in the mismatch between oracle training conditions and predicted inference inputs. Zero-shot evaluation of ChatGPT o4-mini (macro F1: 0.635) and Gemini 3 Fast (macro F1: 0.493) confirms that fine-tuned domain-specific models substantially outperform general-purpose LLMs on this task. We additionally contribute a theoretically grounded tone taxonomy derived from Appraisal Theory, mapping 131 informal annotations to 5 interpretable categories. All models and an interactive demo are publicly available on HuggingFace.

1 Introduction

Natural language understanding requires more than recognising the literal meaning of words. Pragmatics the study of how context shapes meaning is central to social language use. In Burmese, a morphologically and pragmatically complex low-resource language, the same underlying intent can be realised through dozens of distinct utterance forms depending on the social relationship between speakers, the formality of the situation, and the speaker's emotional stance. Correctly identifying these pragmatic properties is a prerequisite for socially aware NLP systems.

Prior work on controllable text generation in Burmese has explored prompting multilingual sequence-to-sequence models with structured attribute inputs (Xue et al., 2021). However, this generation framing presents evaluation challenges: when many surface forms are all valid realisations of the same intent, metrics such as BLEU are unreliable. We argue that a classification framing predicting the pragmatic label of an utterance provides a more tractable and measurable research problem for the same underlying phenomenon.

This paper makes the following contributions:

- We propose a four-stage ablation design for pragmatic classification in Burmese, systematically varying the richness of contextual input to a fine-tuned `xlm-roberta-base` model.
- We introduce three auxiliary classifiers that predict register, power relation, and tone from utterance text, enabling a fully automatic end-to-end pipeline (Stage 4) requiring no manual metadata.
- We quantify the error propagation cost of the pipeline (-0.097 macro F1) and demonstrate that improving intermediate classifier accuracy via a codependent chain or multi-task

architecture does not improve end-to-end performance, identifying the oracle/predicted mismatch as the true bottleneck.

- We evaluate zero-shot LLMs (ChatGPT o4-mini, Gemini 3 Fast) on the same test set with equivalent metadata input, and show that fine-tuned Stage 1 already outperforms both without any metadata.
- We propose a theoretically grounded tone taxonomy grounded in Appraisal Theory (Martin and White, 2005), mapping 131 informal annotations to 5 interpretable categories.
- We release all models and an interactive demo on HuggingFace.

2 Research Questions

This study is structured around four research questions:

RQ1: How much does adding social context and communicative intent to the model input improve Burmese politeness classification compared to an utterance-only baseline?

RQ2: What is the oracle upper bound achievable when explicit pragmatic metadata (register, power relation, tone) is provided as input?

RQ3: How much of the oracle benefit is preserved when pragmatic metadata is predicted automatically, and what is the error propagation cost of the end-to-end pipeline?

RQ4: Can improved intermediate classifiers (codependent chain, multi-task) close the oracle-to-pipeline gap, and how do fine-tuned models compare to zero-shot LLMs?

3 Related Work

3.1 Multilingual Pre-trained Models

XLM-RoBERTa (Conneau et al., 2020) is a cross-lingual encoder trained on 2.5TB of filtered CommonCrawl text across 100 languages, including Burmese. Its encoder-only architecture produces a single contextualised representation that is well-suited for classification tasks. We adopt XLM-RoBERTa-base as our backbone, as it is purpose-built for natural language understanding rather than generation.

The mT5 model (Xue et al., 2021) extends the T5 framework to a massively multilingual setting and has been used for generation-based approaches to Burmese pragmatics. We use mT5 as a baseline point of comparison in framing the task.

3.2 Controlled and Pragmatic NLP

Controllable text generation has explored various mechanisms for conditioning outputs on stylistic or pragmatic attributes (See et al., 2019; Madotto et al., 2020). However, most such work focuses on high-resource languages and coarse attributes such as sentiment or topic. Context-aware dialogue generation has similarly examined politeness and social factors (Li et al., 2016; Rashkin et al., 2019), but rarely in low-resource settings.

For low-resource languages specifically, the interaction between limited data, label quality, and pragmatic complexity presents unique challenges. Our work addresses this by proposing a label coarsening strategy grounded in established sociolinguistic theory.

3.3 Appraisal Theory

Appraisal Theory (Martin and White, 2005) provides a systematic framework for analysing evaluative language. Its Attitude subsystem distinguishes Affect (emotional states), Judgement (social evaluations), and Appreciation (value assessments). We use this framework to map 131 informal tone annotations from our dataset into 5 principled categories, addressing a key annotation quality limitation.

3.4 Error Propagation in NLP Pipelines

Pipeline architectures, where one model's output feeds into the next, are well-known to suffer from error propagation (Finkel et al., 2006). Our Stage 3→Stage 4 analysis contributes a concrete quantification of this effect in the context of pragmatic classification for a low-resource language, and extends it by showing that the bottleneck is not intermediate classifier accuracy per se, but the training/inference distribution mismatch.

4 Dataset

We use the Burmese Contextual Pragmatics dataset (freococo, 2024), a publicly available CC0-licensed resource containing 2,200 Burmese utterances covering 22 root conversational meanings (e.g., "I love you", "Thank you", "Stop it"). Each root meaning has approximately 100 variations realised across different social registers, politeness levels, and situational contexts. An overview of the full system is shown in Figure 1.

Each example provides: the Burmese utterance, a social context description, a communicative in-

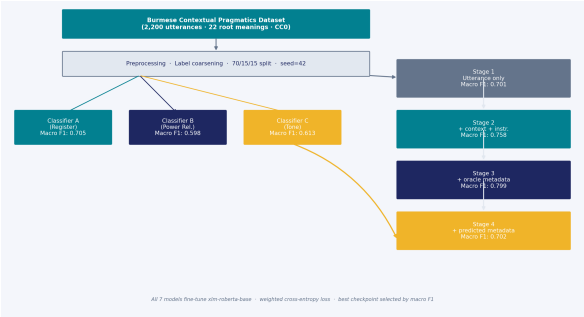

Figure 1: System overview. The dataset feeds into preprocessing and then into models covering Stages 1–4, with three auxiliary classifiers (A, B, C) predicting register, power relation, and tone respectively for the Stage 4 pipeline.

Original label	Mapped to
neutral, polite, informal, professional, blunt, rude	(keep as-is)
very_polite, religious, polite_but_firm	polite
friendly	informal
impolite, sarcastic	rude
stern	blunt

Table 1: Politeness label coarsening: 13 original classes collapsed to 6.

struction, register (colloquial / standard / formal / slang), politeness (13 original labels), power relation (8 original values), tone (131 free-text values), and additional fields including emotion, intent strength, and phonetic transcription.

The dataset was constructed by a single native speaker without a formal annotation protocol, which introduces quality issues particularly in the tone field, which contains typos (e.g., *Appologetic*), compound labels (e.g., *Nervous/Sincere*), and singleton values (e.g., *Spooky*, *Whispering*). We address these issues through principled preprocessing described in Section 5.

5 Preprocessing

5.1 Politeness Label Coarsening

The original politeness field contains 13 labels, several of which have very few instances (e.g., *sarcastic*: 1, *stern*: 2). We merge semantically similar labels into 6 learnable classes using the mapping in Table 1.

5.2 Tone Taxonomy via Appraisal Theory

The tone field presents a more significant challenge: 131 distinct free-text values assigned informally by a single annotator. These are unlearnable as classification targets and lack academic justification.

Category	Representative original values
neutral	Neutral, Direct, Objective, Indifferent
positive	Warm, Sweet, Cheerful, Sincere, Hopeful
formal	Professional, Serious, Respectful, Humble
negative	Angry, Hostile, Harsh, Disgusted, Sarcastic
emotional	Urgent, Pleading, Sad, Panicked, Apologetic

Table 2: Tone taxonomy grounded in Appraisal Theory (Martin and White, 2005).

We address this by mapping the 131 values to 5 categories grounded in the Attitude subsystem of Appraisal Theory (Martin and White, 2005), as shown in Table 2. This is a standard practice in low-resource NLP research where annotation quality is variable (Plank, 2022).

5.3 Power Relation Coarsening

Eight original power relation values are collapsed to four by merging role-specific variants of the inferior relation into a single *inferior_to_superior* class: *customer_to_staff*, *customer_to_seller*, *passenger_to_driver*, and *patient_to_doctor* all reflect the same fundamental speaker hierarchy dynamic.

5.4 Class Imbalance

The politeness label is severely imbalanced: *neutral* constitutes 53% of all examples. Without correction, a model trained with standard cross-entropy loss would learn to predict *neutral* for all inputs, achieving 53% accuracy while failing on all other classes.

We address this with weighted cross-entropy loss, assigning each class a weight of $\frac{N}{K \cdot n_k}$ where N is the total training examples, K is the number of classes, and n_k is the count of class k. Class weights range from 0.318 (neutral) to 6.111 (rude). We report macro F1 as our primary evaluation metric, which treats all classes equally regardless of support.

5.5 Data Splits

We split the dataset 70/15/15 (train/validation/test) using a fixed random seed of 42. The same split is used consistently across all models trained in this work. All reported results are from the held-out test set (330 examples).

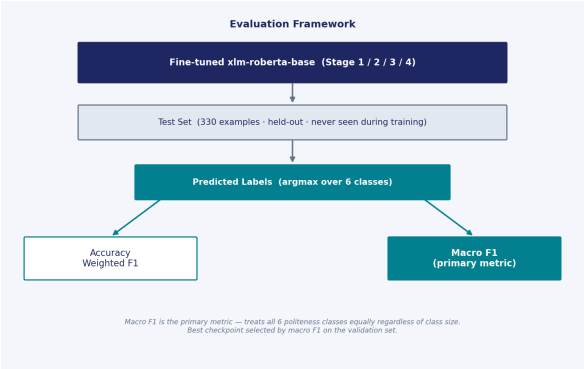

Figure 2: Evaluation framework. All models are evaluated on the held-out test set (330 examples) using accuracy, weighted F1, and macro F1. Macro F1 is the primary metric as it treats all six politeness classes equally regardless of class size.

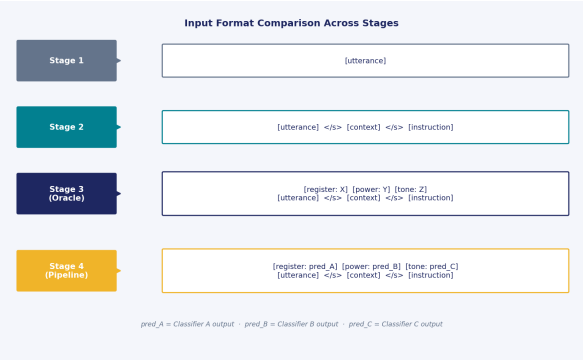

Figure 3: Input format comparison across the four stages. Each stage adds progressively richer contextual information to the model input.

6 Methodology

6.1 Base Model

We fine-tune `xlm-roberta-base` (Conneau et al., 2020) for all classification tasks. XLM-RoBERTa is an encoder-only transformer trained on 100 languages including Burmese, making it appropriate for understanding Burmese pragmatic text. Its classification head consists of a linear layer over the `[CLS]` token representation.

6.2 Training Setup

All models are trained with the following hyperparameters: 8 epochs, batch size 8, learning rate 2×10^{-5}, warmup steps 150, weight decay 0.01, weighted cross-entropy loss. We use HuggingFace `Trainer` with a custom `WeightedTrainer` subclass that overrides `compute_loss` to inject class-weighted loss. The best checkpoint is selected by macro F1 on the validation set using `load_best_model_at_end=True`. The evaluation framework is shown in Figure 2.

6.3 Four-Stage Ablation Design

Our main experimental contribution is a four-stage ablation over the input format given to the politeness classifier. The architecture, training procedure, and evaluation setup are identical across all stages; only the input text changes. This design directly answers RQ1–3.

Stage 1 Utterance only. The model receives only the raw Burmese utterance. This is the baseline, answering whether XLM-RoBERTa can classify Burmese pragmatic properties at all.

Stage 2 Utterance + context + instruction. The social context and communicative instruction are appended as separate segments using the XLM-RoBERTa separator token:

```
[utterance] </s> [context] </s>
[instruction]
```

This tests whether knowing the social situation improves classification.

Stage 3 Oracle metadata prefix. Ground-truth register, power relation, and tone labels are prepended as readable text tags:

```
[register: X] [power: Y] [tone:
Z] [utterance] </s> [context] </s>
[instruction]
```

This is the theoretical upper bound assuming perfect metadata.

Stage 4 Predicted metadata prefix (end-to-end pipeline). The same input format as Stage 3, but metadata values are replaced with predictions from Classifiers A, B, and C respectively. No human-provided metadata is required at inference time.

Figure 3 illustrates the input format used at each stage.

6.4 Auxiliary Classifiers for Stage 4

To enable Stage 4, we train three independently fine-tuned classifiers on the same dataset and split.

Classifier A (Register) takes only the utterance as input and predicts register $\in$ {colloquial, standard, formal, slang}. Register is largely encoded in the text itself through vocabulary and sentence-final particles.

Classifier B (Power relation) takes utterance plus context and predicts power relation $\in$ {equal, inferior_to_superior, superior_to_inferior, any}.

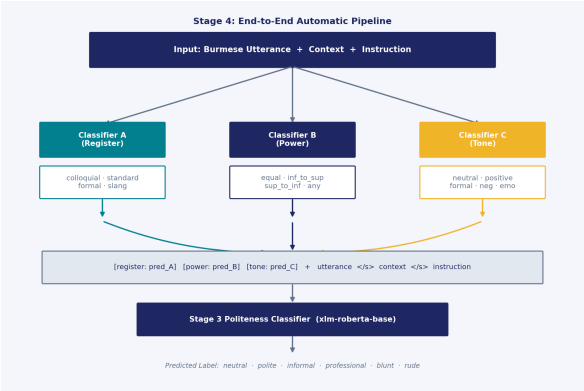

Figure 4: Stage 4 end-to-end pipeline. Three auxiliary classifiers predict register, power relation, and tone automatically; their outputs are assembled into a text prefix that is fed into the Stage 3 politeness classifier.

Context is required because power relation is a social rather than purely linguistic property.

Classifier C (Tone) takes utterance plus context and predicts coarsened tone $\in$ {neutral, positive, formal, negative, emotional}.

Figure 4 shows the full end-to-end flow of the Stage 4 pipeline.

6.5 Improved Intermediate Classifier Architectures

To investigate whether better intermediate classifiers can close the oracle-to-pipeline gap, we explore two alternative architectures.

Codependent Chain (Stage 4 v2). We retrain Classifiers B and C with enriched inputs: Classifier B receives a predicted register prefix (`[register: X]`) prepended to the utterance and context, while Classifier C receives both predicted register and power prefixes. At training time, ground-truth labels are used (teacher forcing); at inference, predictions are chained sequentially A→B→C. This design tests whether register context helps predict power relation, and whether both help predict tone.

Multi-Task Classifier (Stage 4 MT). We train a single XLM-RoBERTa encoder with three independent classification heads one each for register, power relation, and tone jointly optimised with equal-weighted cross-entropy loss. The shared encoder learns representations useful for all three pragmatic dimensions simultaneously. At inference, all three metadata predictions are produced in a single forward pass.

System	Task	Acc.	Macro F1
Original independent classifiers			
Clf. A	Register	0.739	0.705
Clf. B	Power relation	0.739	0.598
Clf. C	Tone	0.667	0.613
Codependent chain (v2)			
Clf. B_v2	Power relation	0.770	0.637
Clf. C_v2	Tone	0.655	0.621
Multi-task (shared encoder)			
MT head	Register	0.787	0.719
MT head	Power relation	0.787	0.660
MT head	Tone	0.677	0.632

Table 3: Intermediate classifier results. All variants improve over the original independent classifiers, with the multi-task model achieving the highest scores across all three tasks.

7 Results

7.1 Intermediate Classifiers

Table 3 reports test-set results for all intermediate classifier variants.

Classifier A achieves the strongest performance among the original classifiers. *colloquial* (F1=0.82) and *formal* (F1=0.79) are well-separated, but *standard* (F1=0.53) remains difficult due to its ambiguous middle-ground position between registers. Classifier B struggles with *any* (F1=0.40) and *superior_to_inferior* (F1=0.43), both of which are underrepresented and semantically difficult. Classifier C achieves moderate performance; *negative* (F1=0.41) has only 17 test examples.

The codependent chain improves both B and C: B_v2 gains +0.039 and C_v2 gains +0.008. The multi-task model achieves the strongest results across all heads, with power relation improving most substantially (+0.062 over the original independent classifier).

7.2 Politeness Classification: Main Ablation

Table 4 presents the full ablation results including all Stage 4 variants and zero-shot LLM baselines.

7.3 Per-Class Results

Table 5 shows per-class F1 scores for Stage 3, Stage 4, and the two zero-shot LLMs.

8 Analysis

RQ1: Does context help? Yes. Stage 2 improves over Stage 1 by +0.057 macro F1, confirming that social context and communicative instruction carry real pragmatic signal. The utterance

Model	Acc.	Macro F1
Fine-tuned ablation		
Stage 1 utterance only	0.739	0.701
Stage 2 + context + instruction	0.794	0.758
Stage 3 oracle metadata	0.846	**0.799**
Stage 4 predicted (independent)	0.727	0.702
Stage 4 v2 codependent chain	0.724	0.696
Stage 4 MT multi-task	0.773	0.681
Zero-shot LLM baselines (oracle metadata input)		
ChatGPT o4-mini	0.564	0.635
Gemini 3 Fast	0.500	0.493

Table 4: Full results. Macro F1 is the primary metric. Zero-shot LLMs receive the same oracle metadata as Stage 3 for a fair comparison. Stage 1 outperforms both zero-shot LLMs despite receiving no metadata.

Class	Stage 3	Stage 4	ChatGPT	Gemini
blunt	0.67	0.70	0.47	0.42
informal	0.83	0.71	0.59	0.51
neutral	0.87	0.77	0.45	0.54
polite	0.82	0.67	0.64	0.36
professional	0.94	0.91	0.79	0.52
rude	0.67	0.46	0.87	0.60
macro avg	0.80	0.70	0.64	0.49

Table 5: Per-class F1 comparison across models.

alone is insufficient for reliable politeness classification, which is linguistically expected: Burmese politeness marking depends heavily on speaker-listener relationship and situational formality.

RQ2: What is the oracle upper bound? Stage 3 achieves 0.799 macro F1, a further +0.041 over Stage 2. When the model is given exact register, power relation, and tone labels, its classification improves meaningfully. This demonstrates that these pragmatic dimensions are genuinely informative for politeness prediction.

RQ3: How much does error propagation cost? Stage 4 drops to 0.702 macro F1, -0.097 below the oracle (Stage 3). Strikingly, Stage 4 achieves nearly identical performance to Stage 1 (0.701), despite having access to predicted metadata. This demonstrates that *noisy predicted metadata is almost as uninformative as no metadata at all*. The bottleneck is intermediate classifier accuracy: Classifiers A, B, and C achieve macro F1 of 0.705, 0.598, and 0.613 respectively, and their errors negate the metadata benefit identified in Stage 3.

RQ4: Do better intermediate classifiers help, and how do LLMs compare? No and this is the most surprising finding of the paper. Both improved architectures the codependent chain (0.696) and multi-task model (0.681) perform *worse* than the original independent classifiers (0.702) at the Stage 4 level, despite having measurably better intermediate classifiers. The ordering of Stage 4 performance is exactly the reverse of intermediate classifier quality:

$$\text{Original } (0.702) > \text{Codependent } (0.696) > \text{Multi-task } (0.681)$$

Yet intermediate quality goes:

$$\text{Multi-task} > \text{Codependent} > \text{Original}$$

This reveals that the true bottleneck is not intermediate classifier accuracy but the *distribution mismatch* between training and inference conditions. The Stage 3 politeness model was trained exclusively on oracle (ground-truth) metadata prefixes. At inference, even improved predicted prefixes carry a different error distribution than what the model trained on. The original independent classifiers, despite being weakest, produce predictions that happen to be most compatible with the distribution the Stage 3 model learned from. Better classifiers make qualitatively different errors that the politeness model handles less gracefully.

For zero-shot LLMs, fine-tuned Stage 1 (utterance only, no metadata, 0.701) outperforms ChatGPT o4-mini (0.635) and Gemini 3 Fast (0.493), even though the LLMs received full oracle metadata. This gap is consistent with Burmese being under-represented in general LLM training data: pragmatic classification in a low-resource language requires domain-specific fine-tuning that prompting alone cannot substitute.

Value of the tone taxonomy. The Appraisal Theory mapping is an enabling contribution: without reducing 131 annotations to 5 principled categories, Classifier C could not be trained, and Stage 4 would be impossible. The mapping also makes the tone labels interpretable and academically citable, addressing a key weakness of the original dataset.

9 Error Analysis

9.1 Stage 3 vs. Stage 4 Disagreements

To understand the error propagation mechanism qualitatively, we identify the 49 test examples where Stage 3 (oracle metadata) predicts correctly but Stage 4 (predicted metadata) does not. We

analyse metadata errors and their downstream effects on politeness classification.

Metadata error rates. On the 330 test examples, register errors occur in 26.1% of cases, power relation errors in 26.1%, and tone errors in 33.0%. However, their impact on final politeness classification is uneven. Among the 49 disagreement cases, register errors are present in 63.3%, tone errors in 61.2%, and power errors in 46.9%, suggesting that register and tone errors are disproportionately harmful relative to their raw frequency.

The *neutral* → *polite* confusion dominates. The most frequent error pattern is the misclassification of neutral utterances as polite (21/49 cases). This consistently arises when Classifier A upgrades the register from *colloquial* to *standard*. For example, the utterance " " (Thank you for the birthday wish) in a casual peer context is correctly identified as *colloquial* by the oracle, yielding a *neutral* prediction. However, Classifier A predicts *standard* register, which causes the Stage 3 model to predict *polite* an inflated politeness level inconsistent with the social context.

Power relation errors inflate politeness predictions. A secondary pattern involves misclassification of equal power relations as *inferior_to_superior*. When the model incorrectly infers a hierarchical relationship, it systematically predicts *polite* for utterances that are contextually *neutral*. For instance, the exclamation "... " (upon touching something hot) is correctly labelled *neutral* under oracle metadata, but predicted as *polite* when power is misclassified as *inferior_to_superior*.

Tone errors cause bidirectional confusions. Tone misclassification produces more varied errors. The *neutral* → *negative* → *rude* pattern illustrates under-prediction: a helpful but ignorant response is misclassified with negative tone and predicted as *rude*. Conversely, *neutral* → *emotional* leads to the *neutral* → *informal* confusion. These bidirectional errors confirm that tone is the most semantically ambiguous dimension for the intermediate classifier.

9.2 Per-Class Performance Across Models

Professional is the most consistently learnable class across all models (Stage 3: 0.94, Stage 4: 0.91, ChatGPT: 0.79), reflecting the distinctiveness of formal Burmese workplace vocabulary and particles. Even zero-shot LLMs handle this class reasonably well, suggesting that professional register carries strong surface-level signals.

Neutral shows the sharpest contrast between fine-tuned and zero-shot models. Stage 3 achieves F1 0.87 on *neutral*, while ChatGPT drops to 0.45 and Gemini to 0.54. Both LLMs massively under-predict neutral (ChatGPT: 50 predictions vs. 174 true; Gemini: 103 vs. 174), deflecting to *informal* and *polite* instead. Fine-tuning with weighted cross-entropy directly addresses this imbalance, explaining why even Stage 1 substantially outperforms zero-shot LLMs.

Rude shows an interesting reversal: ChatGPT achieves F1 0.87, higher than Stage 3 (0.67). This suggests explicit rudeness markers in Burmese are detectable by general LLMs, though the small support (12 examples) warrants caution in interpreting this result.

9.3 Zero-Shot LLM Failure Patterns

Both LLMs exhibit a systematic bias toward *informal* over-prediction. ChatGPT predicts *informal* 131 times against a true count of 59 (+72 over-prediction), while Gemini predicts it 120 times (+61). The dominant confusion in both models is *neutral* → *informal*: 66 cases in ChatGPT and 56 in Gemini. This suggests that general LLMs lack the sociolinguistic grounding to distinguish pragmatic neutrality from casual register in Burmese a distinction that depends on speaker relationship and situational context rather than lexical content alone.

A secondary shared failure is *neutral* → *blunt* (22 cases in ChatGPT, 18 in Gemini). Utterances that are factual and direct but not impolite are systematically interpreted as blunt, indicating over-indexing on directness as a bluntness marker.

The most important finding from the zero-shot analysis is the *neutral* class collapse: both LLMs predict *neutral* far less than its true frequency. Since *neutral* constitutes 53% of the dataset, this systematic under-prediction is the primary driver of low macro F1 scores. This is not fixable by prompt engineering alone it requires exposure to the distributional properties of the specific dataset, which only fine-tuning provides.

10 Direction Change: From Generation to Classification

This project originally proposed a generation framing: given structured pragmatic attributes, generate the appropriate Burmese utterance us-

ing mT5. We pivoted to classification for three reasons. First, the dataset contains only 22 root meanings with approximately 100 near-identical surface variations each, providing insufficient diversity for a generation model to generalise. Second, BLEU (Papineni et al., 2002) and chrF (Popović, 2015) are unreliable when multiple surface forms are valid realisations of the same intent. Third, early training runs confirmed non-convergence: training loss remained at 27+ over 8 epochs, and one run produced `training_loss=0.0`, `validation_loss=NaN` due to a GPU FP16 precision issue.

Classification with XLM-RoBERTa provides falsifiable research claims, clean metrics, and a well-defined evaluation setup. The research questions shift from "can the model generate appropriate utterances?" to "can the model identify pragmatic properties of utterances?", which is an equally valid and more measurable NLP problem for this dataset.

11 Limitations

This study has several limitations. The dataset is small (2,200 examples) and covers only 22 root meanings, which constrains generalisability. The tone taxonomy, while theoretically grounded, was applied manually and a future annotation study could verify inter-annotator agreement on the coarsened categories. The evaluation of the end-to-end pipeline is limited to a single test split; cross-validation would provide more stable estimates of the error propagation cost. The zero-shot LLM evaluation uses a single prompt design; prompt engineering may yield different results. Finally, all experiments use a single base model (`xlm-roberta-base`); larger models or Burmese-specific pre-trained models may yield different trade-offs.

12 Conclusion

We have presented a four-stage ablation study for Burmese pragmatic politeness classification, extending from a simple utterance-only baseline to a fully end-to-end pipeline with automatic metadata prediction. Our results show that social context is a meaningful signal (+0.057 macro F1), oracle pragmatic metadata provides a further gain (+0.041), and the end-to-end pipeline incurs a significant error propagation cost (-0.097) that nearly eliminates the metadata benefit.

A key finding of this work is that improving intermediate classifier accuracy through codependent training or multi-task learning does not improve end-to-end Stage 4 performance. All three Stage 4 variants cluster near the Stage 1 baseline (0.696–0.702), despite intermediate classifier macro F1 improvements of up to +0.062. This reveals that the bottleneck is not classifier accuracy but the oracle/predicted distribution mismatch: the Stage 3 politeness model was trained on clean metadata prefixes and cannot fully exploit noisily predicted ones. Future work should explore co-training the politeness model jointly with the metadata predictors to close this gap.

Zero-shot evaluation confirms that fine-tuning on domain-specific Burmese pragmatic data is essential: fine-tuned Stage 1 (utterance only) outperforms ChatGPT o4-mini with full oracle metadata, and substantially outperforms Gemini 3 Fast. The systematic *neutral* class collapse in both LLMs predicting neutral far less than its 53% true frequency illustrates that general LLMs lack the distributional grounding for Burmese pragmatic classification that fine-tuning provides.

We contribute a theoretically grounded tone taxonomy and release all models publicly. Future work will focus on co-training the full pipeline end-to-end and exploring Burmese-specific pre-trained encoders.

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